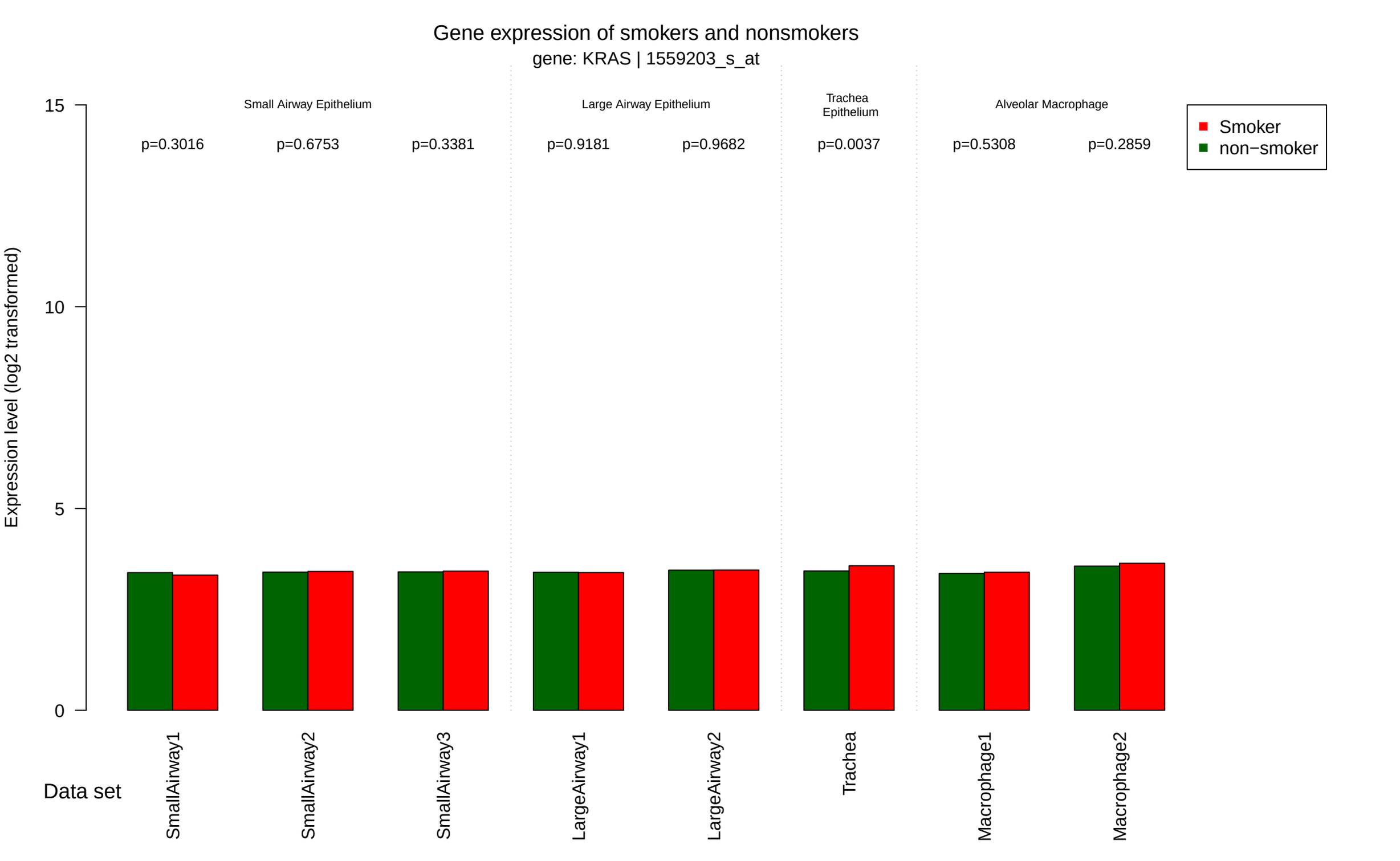
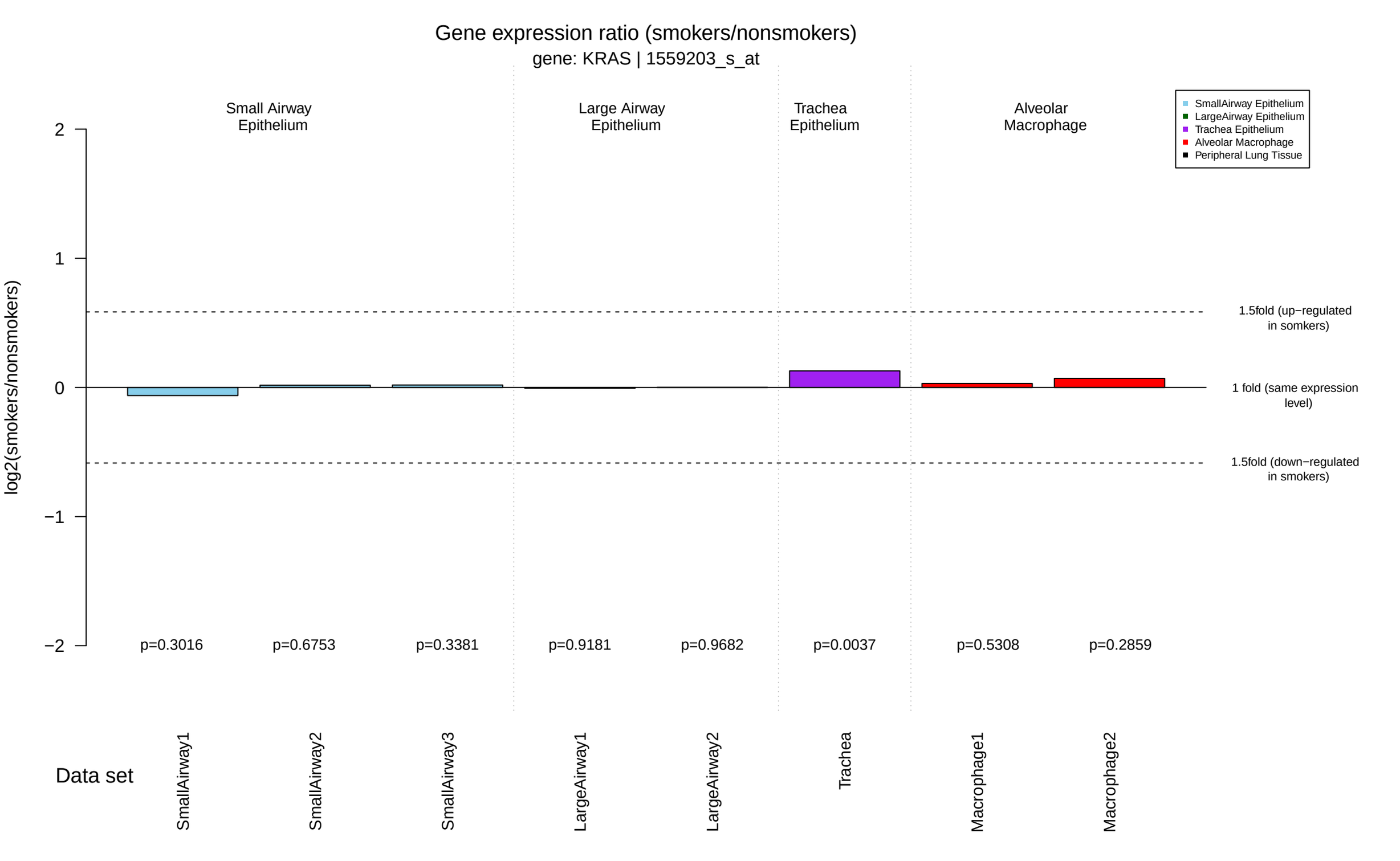
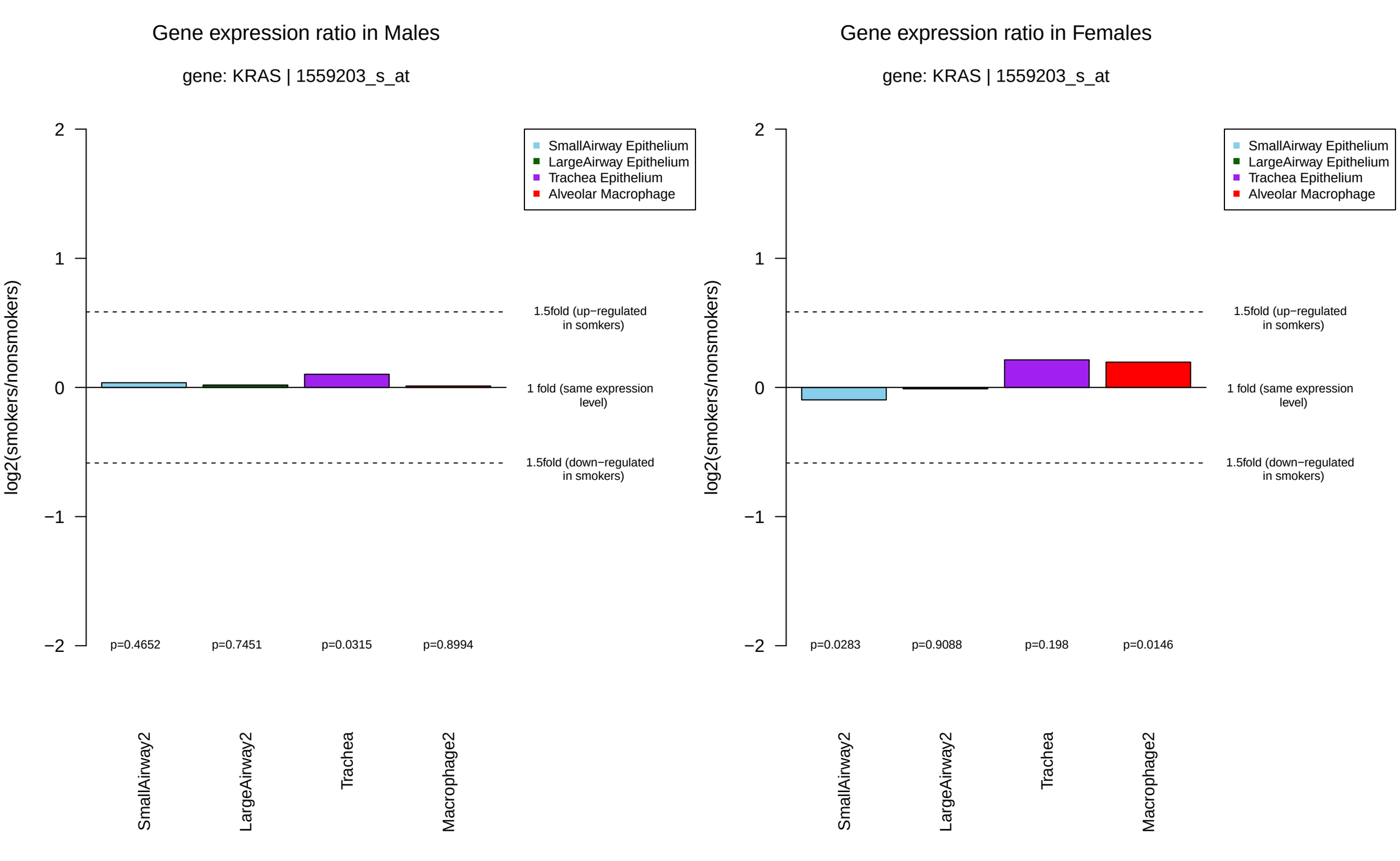
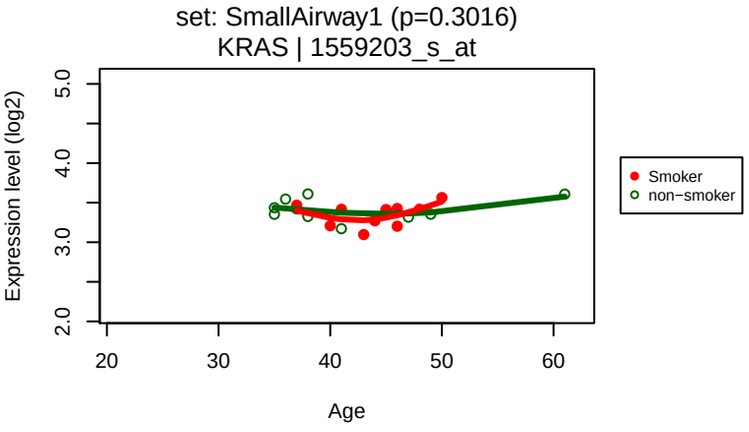
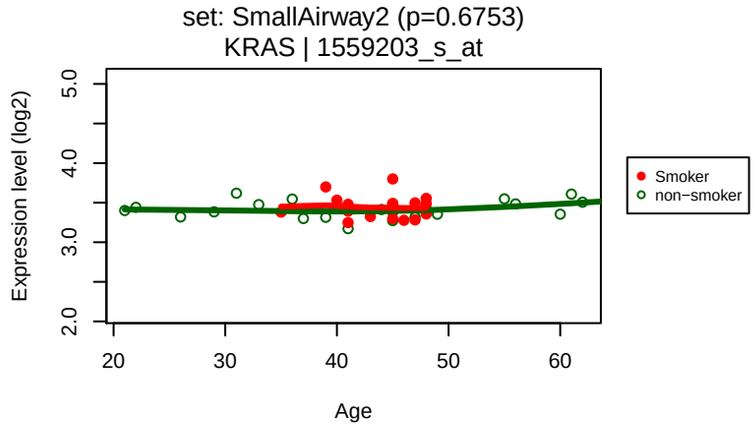
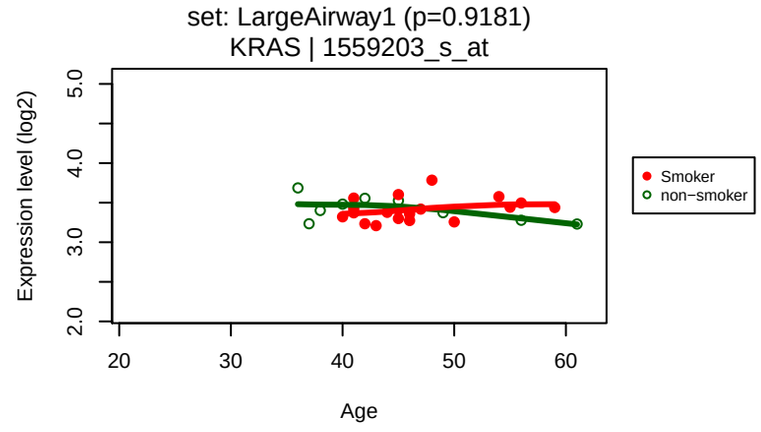

Gene expression of smokers and nonsmokers
gene: KRAS | 1559203_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3016
p=0.6753
p=0.3381
p=0.9181
p=0.9682
p=0.0037
p=0.5308
p=0.2859
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

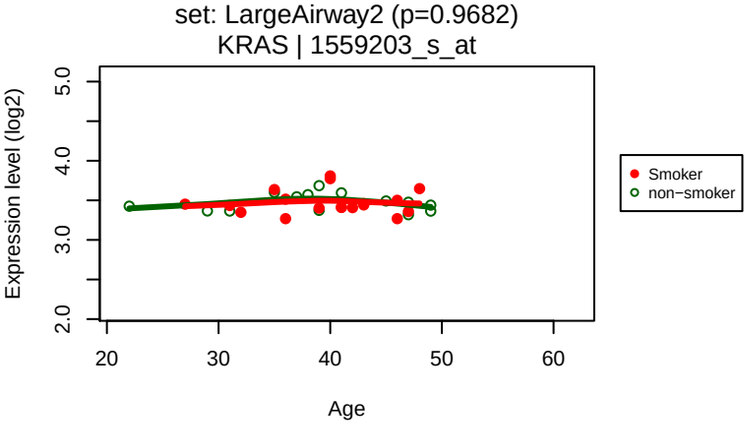

Gene expression ratio (smokers/nonsmokers)
gene: KRAS | 1559203_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3016
p=0.6753
p=0.3381
p=0.9181
p=0.9682
p=0.0037
p=0.5308
p=0.2859
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

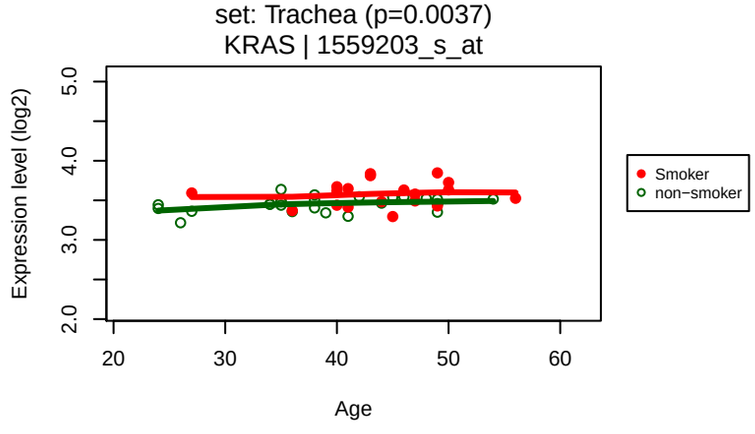

Gene expression ratio in Males
gene: KRAS | 1559203_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4652
p=0.7451
p=0.0315
p=0.8994
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KRAS | 1559203_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0283
p=0.9088
p=0.198
p=0.0146
SmallAirway2
LargeAirway2
Trachea
Macrophage2

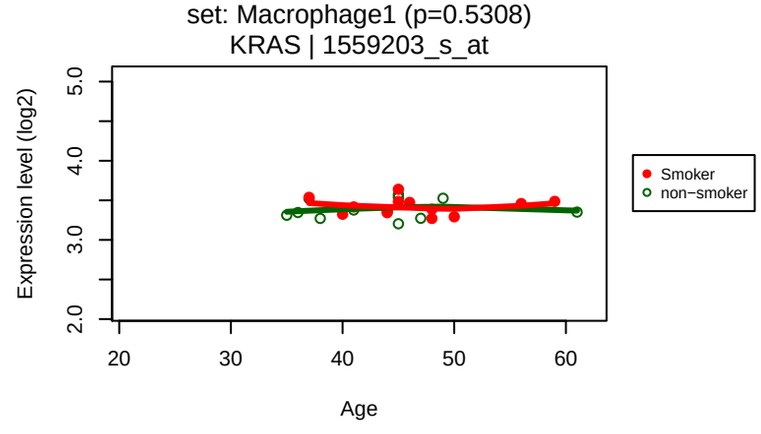

set: SmallAirway1 (p=0.3016)
KRAS | 1559203_s_at
Expression level (log2)
Age
Smoker
non−smoker

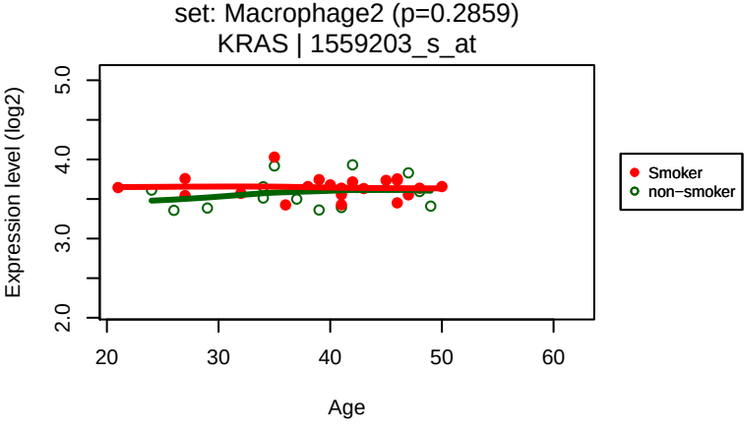

set: SmallAirway2 (p=0.6753)
KRAS | 1559203_s_at
Expression level (log2)
Age
Smoker
non−smoker

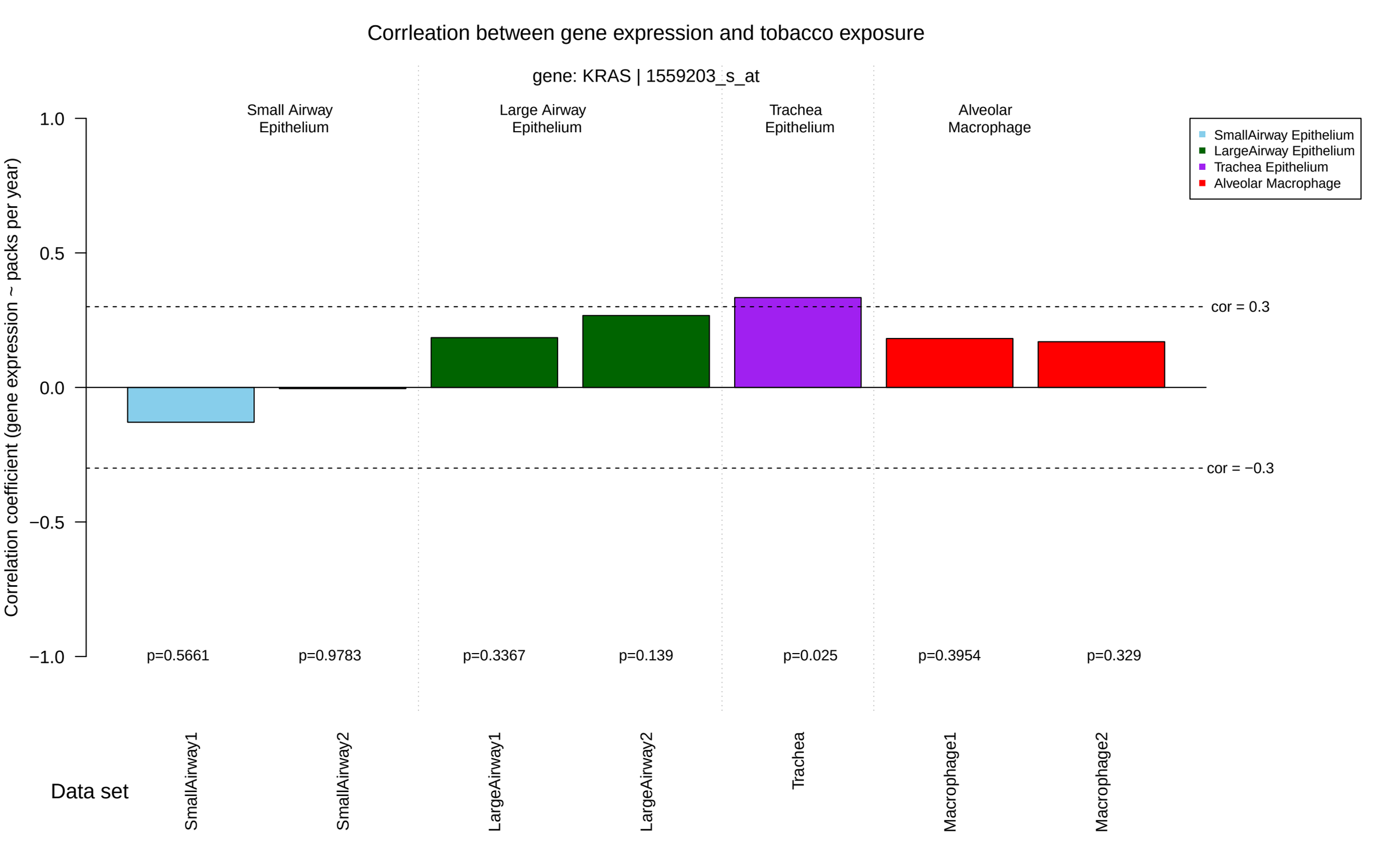

set: LargeAirway1 (p=0.9181)
KRAS | 1559203_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9682)
KRAS | 1559203_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0037)
KRAS | 1559203_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5308)
KRAS | 1559203_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2859)
KRAS | 1559203_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KRAS | 1559203_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5661
p=0.9783
p=0.3367
p=0.139
p=0.025
p=0.3954
p=0.329
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2